

Improved methanol yield and selectivity from CO₂ hydrogenation using a novel Cu-ZnO-ZrO₂ catalyst supported on Mg-Al layered double hydroxide (LDH)



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ARTICLE INFO

Keywords:

Cu-ZnO-ZrO₂
LDH
Co-precipitation
CO₂ hydrogenation
Methanol

ABSTRACT

Methanol synthesis via CO₂ hydrogenation is an important part of the strategy for generating clean energy as we attempt to reduce our dependency on fossil fuels. Conventional catalysts for this reaction need improvement in their methanol selectivity. In this work, a layered double hydroxide (Mg-Al LDH) was used as a carrier for Cu-ZnO-ZrO₂ to produce a catalyst by co-precipitation. From characterization results, CuO-ZnO-ZrO₂ nanoparticles were formed and were uniformly dispersed and attached to the surface of LDH. BET surface area and copper dispersion of the catalysts were significantly improved by 4.3 times and 2.9 times, respectively, compared with a reference catalyst without the support. In a catalytic reaction, the catalyst showed dramatic methanol selectivity of 78.3% at 523 K and 3.0 MPa, which is 14.4% higher than the commercial catalyst measured in this investigation and about 50% higher than conventional copper-based catalysts in literatures. It also showed over twice the space time yield based on active metal sites compared to a commercial catalyst in the temperature range 473 K–573 K. Therefore, the prepared catalyst can be efficiently applied at relatively mild reaction temperatures and pressures.

1. Introduction

Using carbon dioxide (CO₂) as a carbon source to produce platform chemicals has stimulated great interest as a part of the efforts to mitigate CO₂ emissions and provide clean energy. Renewable fuels generated from CO₂ can serve as a substitute for conventional fossil fuels which have been used for centuries. Since the large-scale application of fossil fuels is the dominant contributor to elevated atmospheric CO₂ concentration, a sustainable carbon cycle involving CO₂ utilization is expected to contribute to easing the current crisis on climate and energy [1–3]. Methanol (MeOH) synthesis, using CO₂ captured from CO₂ emission sources, is one of the most attractive options in CO₂ utilization [4]. Motivated by the continually reduced hydrogen manufacturing costs, hydrogen generated from renewable energy can be reacted with CO₂ to produce methanol, which is transported and stored far more readily than hydrogen alone. The methanol produced in this way can be applied widely in the chemical industry, as a solvent or raw material, or used in fuel cells directly [5,6].

The process of CO₂ hydrogenation, however, relies heavily on an efficient catalyst. Apart from the numerous catalysts containing noble or rare metals [7,8], copper based catalysts are the most important type for this reaction due to their excellent reactivity and low cost. Catalysts CuO-ZnO and further developed CuO-ZnO-Al₂O₃ play crucial roles in methanol manufacturing today [9,10]. While effective for conversion of syngas to methanol, the current copper-based commercial catalysts experience serious shortcomings when CO₂ replaces CO, because CO₂ is more inert than CO, leading to lower CO₂ conversion. Furthermore, water produced during the reactions results in sintering of the active copper [10,11]. Recently, incorporation of zirconium (Zr) into the traditional Cu-Zn catalyst system has been undertaken because zirconium (Zr) promotes the formation of smaller copper particles, provides higher copper dispersion [12], and also helps stabilize active copper by preventing agglomeration during synthesis and sintering during reaction [13]. These apparent benefits come at a cost, however. Most CuO-ZnO-ZrO₂ catalysts possess a low surface area of only 30 m² g^{−1}, which limits its catalytic function in the reaction. Arena and co-workers

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<https://doi.org/10.1016/j.jcou.2018.11.006>

Received 31 August 2018; Received in revised form 1 November 2018; Accepted 15 November 2018

Available online 28 November 2018

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developed a novel route possessing to prepare catalysts, and obtained large surface area and well copper dispersed Cu-ZnO/ZrO₂ samples [14].

Accordingly, much effort has been devoted to enhancing the catalytic properties of CuO-ZnO-ZrO₂ by dispersing it on porous carriers such as silica, alumina, carbon nanotubes, etc. [15–17]. One especially attractive candidate support material is layered double hydroxides (LDHs) for reasons discussed later, but there are very few systematic studies using layered double hydroxides (LDHs). LDHs consist of metal hydroxide (positively charged) and interlayers (negatively charged), and have great potential as a catalyst carrier [18]. Gao et al. synthesized CuO-ZnO-ZrO₂ catalysts via hydrotalcite-like precursors in their pioneering work to elevate the catalytic properties in aspects of CO₂ conversion and methanol selectivity [8,19,20], confirming the superiorities of Cu-ZnO-ZrO₂ catalyst derived from LDH precursors on CO₂ hydrogenation. We show in this work that LDHs (employed as a carrier) can provide considerable surface area of approximately 200 m² g⁻¹ for catalyst dispersion, and its favorable sorption capabilities for CO₂ and H₂O at high temperatures [21–23] can elevate the catalytic performance at low CO₂ partial pressures and remove produced H₂O molecules during CO₂ hydrogenation. Thus, these features appear as a “sorption enhanced” effect [24,25] during methanol synthesis, which is expected to greatly facilitate performance of the catalyst, especially at mild reaction conditions.

In this work, a novel synthesis procedure was used to anchor CuO-ZnO-ZrO₂ nanoparticles (Cu:Zn:Zr mole ratio of 6:3:1 [26,27]) onto Mg-Al LDH by co-precipitation. Topologies and morphologies of the as-obtained heterogeneous catalysts were analyzed with the help of characterization including XRD, SEM-EDX, N₂ physisorption, etc. Catalytic properties of the novel catalysts were investigated in a fixed-bed micro reactor.

2. Materials and methods

2.1. Preparation of Cu-ZnO-ZrO₂ catalyst without support

Cu-ZnO-ZrO₂ catalyst without support was first synthesized for comparison with the LDH supported materials. They were prepared via a strategy developed from conventional co-precipitation methods. Typically, a specified amount of Cu(NO₃)₂·2.5H₂O, Zn(NO₃)₂·6H₂O and ZrO(NO₃)₂·xH₂O (Sigma-Aldrich) was dissolved in distilled water according to a mole ratio of Cu:Zn:Zr = 6:3:1 to prepare 1 mol L⁻¹ metal nitrate solution. Then 1.2 mol L⁻¹ K₂CO₃ (Ajax) solution was added to the metal nitrate solution under vigorous agitation at ambient temperature until the pH reached 9.5, following by 2 h stirring and 1 h aging. After washing, centrifuging and drying at 393 K overnight, the precipitate was calcined at 673 K for 4 h in air – this sample was denoted as CZZ. The CZZ was activated *in situ* (aCZZ) at 573 K for 6 h in a pure hydrogen flow before reaction.

2.2. Preparation of Cu-ZnO-ZrO₂/LDH heterogeneous catalysts

CuO-ZnO-ZrO₂ supported on LDH heterogeneous catalyst was synthesized following the improved co-precipitation method involving ultrasonic irradiation. The commercial layered double hydroxide (LDH) Pural MG50 (SASOL Germany GmbH), as the support, was activated at 673 K for 4 h in air (labeled as aLDH). In a typical procedure, 6 g aLDH were evenly distributed in 25 mL of 1 mol L⁻¹ trimetallic nitrate solution (Cu: Zn: Zr = 6: 3: 1 on a mole basis) to obtain a uniform suspension. The samples were then ultrasonicated for 5 min and stirred at ambient temperature for 2 h. K₂CO₃ solution at a concentration of 1.2 mol L⁻¹ was then added drop-wise to the suspension under vigorous stirring to adjust the pH to approximately 9.5. Under vigorous agitation, the solution was stirred for another 2 h and aged for 1 h. The precursors were gathered by centrifuge, washed with distilled water, and dried at 393 K overnight (labelled as pCZZ-LDH). The sample was

calcined at 673 K for 4 h to obtain the resultant CuO-ZnO-ZrO₂/LDH catalyst (CZZ-LDH). Finally, the CZZ-LDH was reduced *in situ* (aCZZ-LDH) at the same conditions as CZZ.

2.3. Material characterization

The crystal phases of the samples were measured by powder X-ray diffraction (PXRD, PW1140/90, Phillips Analytical) equipped with Cu-Kα radiation (2θ from 8 to 70°). Morphologies and chemical compositions of the samples were characterized by high resolution scanning electron microscope and energy dispersive spectroscopy (SEM-EDX, JSM-7001 F, JEOL), respectively. Bulk chemical compositions were measured with the inductive coupled plasma emission spectrometer (ICP-OES, Varian 720ES). Surface areas and pore size distributions were calculated by Brunauer-Emmett-Teller (BET) model and Barrett-Joyner-Halenda (BJH) model based on N₂ isotherms at 77 K (BELSORP-max, MicrotracBEL). Before measurement, the sample was degassed at 373 K for 1 h and then at 573 K for another 3 h until vacuum pressure was lower than 10⁻² kPa. Copper surface area and copper dispersion were analyzed by the N₂O titration method (BELCAT-M, MicrotracBEL) [28]. In a typical measurement process, 0.1 g of catalyst was filled in a quartz U-tube and then the tube was installed in the apparatus equipped with a mass spectrometry. The temperature was slowly raised from ambient temperature to 973 K in 5% H₂ (balanced with Ar, Coregas) flow (5 K min⁻¹) and held at 973 K for 2 h until the catalyst was fully reduced. After cooling to 333 K in Ar flow, it was exposed to 5% N₂O (balanced with He, Coregas) flow for 1 h at 333 K so that Cu⁰ in the surface was oxidized into Cu₂O. The catalyst was again reduced in 5% H₂ (balanced with Ar, Coregas) flow by repeating the reduction process above. H₂ consumption was recorded by mass spectrometry during the entire measurement. In the calculation, the reaction of copper oxidization is assumed in Cu/N₂O mole stoichiometry of 2:1 and an atomic Cu surface density is 1.49 × 10¹⁹ atoms/m² [13,29,30]. The micro structure of the reduced aCZZ-LDH was analyzed with helium ion microscopy (HIM; ORION NanoFab, Carl Zeiss).

2.4. Catalytic property analysis

Catalytic reaction experiments were conducted in a fixed-bed micro reactor (column ID 10 mm x L 500 mm) as shown in Fig. 1. The reactor included three sections: the gas preparation and delivering system, the reaction column, and the sample collection. In the inlet gas system,

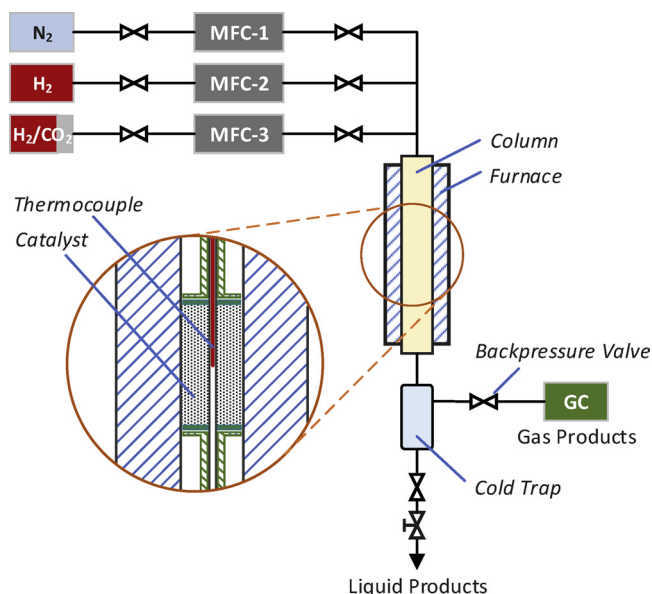


Fig. 1. Schematic of the fixed-bed micro reactor.

three mass flow controllers were used to maintain feed gas flowrate, where N₂ was used to purge the whole system. In the reaction column, the effective reaction height was 4 cm so that the maximum catalyst volume is 3.14 cm³; the maximum reaction temperature can be controlled to 673 K by a temperature controller and the gas pressure during reaction is maintained with a back pressure regulator. At the exit of reactor, gas and liquid products were separated by a cold trap, so that the gas product sample can be collected after the backpressure regulator and the liquid product collected in a liquid separation vessel.

2.4.1. Pre-treatment

Before catalytic reaction, the prepared catalyst (CZZ-LDH or CZZ) was reduced. The mixture of 1 g of tableted catalyst (diameter within 200–500 μm) and 1 g of quartz sand (50–70 mesh, Sigma-Aldrich) was loaded into the centre of the reaction column for pre-treatment, where the catalyst was reduced *in situ* in pure H₂ flow at 30 cm³ (STP) min^{−1} (maintained with MFC-2). From our previous experimental results of H₂ temperature programmed reduction (H₂-TPR) and reductions in both of U-tube and reactor, the prepared catalyst can be completely reduced at 573 K and 6 h. Thus, the temperature of 573 K and the reduction time of 6 h were set in our reduction experiments. After reduction, the catalysts were cooled to ambient temperature in a flow of nitrogen.

2.4.2. Catalytic reaction and measurement

A mixed gas of H₂ and CO₂ (74.98% H₂ and 25.02% CO₂, Coregas) was fed into the reaction system at a specified flow rate maintained with a mass flowmeter (MFC-3), and the gas pressure in the system was fixed at 3.0 MPa by the backpressure regulator. The reaction reached steady-state after 4 h by identifying the CO₂ conversion and methanol selectivity in the products. The outlet flowrate was then measured and recorded once every two hours with a bubble flowmeter, and the gas and liquid products were collected and quantitatively analyzed with a gas chromatograph (GC7890B, Agilent) equipped with thermal conductivity detector (TCD) and flame ionization detector (FID). In the gas products, CO and CH₄ were monitored. After each experiment, the mass and volume of mixture of catalyst and quartz sand were measured.

2.4.3. Data analysis

CO₂ conversion (X_{CO₂}) and methanol selectivity (S_{MeOH}) were calculated as

$$X_{CO_2} = \left[1 - \frac{F_{out} \times p_{CO_2}}{F_{in} \times 25.02\%} \right] \times 100\% \quad (1)$$

$$S_{MeOH} = \left[1 - \frac{F_{out} \times (p_{CO} + p_{CH_4})}{F_{in} \times 25.02\% - F_{out} \times p_{CO_2}} \right] \times 100\% \quad (2)$$

where F_{in} was inlet flowrate (smL min^{−1}), F_{out} was outlet flowrate (smL min^{−1}), p_{CO_2} , p_{CO} and p_{CH_4} were molar percentages (%) of CO₂, CO and CH₄ in the off-gas. Since only by-products of CO and CH₄ were detected in the products, a carbon balance was calculated based on CO₂, methanol, CO and CH₄ in the present work.

Two frequently-used parameters viz. gaseous hourly space velocity (GHSV, h^{−1}) and space time yield (STY_{total}, mg_{MeOH} g_{catalyst}^{−1} h^{−1}) were employed to compare catalyst efficiencies – these are useful parameters since they are scale independent.

$$GHSV = \frac{F'_{in}}{V} \quad (3)$$

where F'_{in} (mL/h) was the hourly volumetric flowrate of feed gas and V was the catalyst volume (including quartz sand).

$$STY_{total} = \frac{m_{MeOH}}{m_{total\ catalyst} \times t} \quad (4)$$

Another two space time yields viz. STY_{active component} and STY_{Cu} were calculated on the mass basis of active component as

$$STY_{active\ component} = \frac{m_{MeOH}}{m_{active\ component} \times t} = \frac{STY_{total}}{p_{active\ component}} \quad (5)$$

$$STY_{Cu} = \frac{m_{MeOH}}{m_{Cu} \times t} = \frac{STY_{total}}{p_{Cu}} \quad (6)$$

where STY_{active component} (mg_{MeOH} g_{active component}^{−1} h^{−1}) was the mass of methanol generated per gram active component per hour, STY_{Cu} (mg_{MeOH} g_{Cu}^{−1} h^{−1}) was the mass of methanol generated per gram copper per hour, m_{MeOH} (g) was the mass of methanol produced, m_{total} (g) was the total mass of catalyst, $m_{active\ component}$ (g) was the mass of active component in the catalyst, m_{Cu} (g) was the mass of copper in the catalyst, t (h) was the period of reaction, $p_{active\ component}$ (wt.%) was the weight percentage of metal catalyst viz. Cu-ZnO-ZrO₂, and p_{Cu} (wt.%) was the weight percentage of copper. As metal catalyst is the most important active component in catalysts, the STY_{active component} and STY_{Cu} can help better compare catalytic performances without the influence of the LDH carrier.

3. Results and discussion

3.1. Characterization

3.1.1. Textural properties

The crystallinity and structure for the different samples at different levels of preparation were observed by XRD and are shown in Fig. 2. Only the sharp diffraction peaks of layered double hydroxide can be observed in LDH [31], reflecting its double layered microstructure with a layer spacing of $d_{003} = 0.768$ nm. The aLDH (activated LDH) shows three broad peaks at $2\theta = 35, 44$ and 62° as expected because the crystalline LDH was partly converted to an amorphous state due to dehydration and decarbonization. Herein, interlayer anions (hydroxyl and carbonate) in LDH decompose simultaneously at 673 K [32]. In the pattern of pCZZ-LDH, apart from the peaks of Cu₂(OH)₃NO₃ (PDF# 14-0687) which are common in co-precipitation precursors, the detectable peak at $2\theta = 11.7^\circ$ signals the reappearance of a double layered structure with a tiny shift in layer space ($d_{003} = 0.764$ nm) because of a “memory effect” [33–35]. When the catalyst precursor pCZZ-LDH was calcined, the Cu₂(OH)₃NO₃ decomposed at 673 K and the remaining crystal structure was further destroyed, thus, CZZ-LDH only shows diffraction peaks of CuO, ZnO and amorphous LDH (see Figure S1 in SI for XRD of CZZ). After reduction, aCZZ-LDH pattern shows peaks of Cu, ZnO and amorphous LDH, demonstrating that active sites of Cu⁺ are

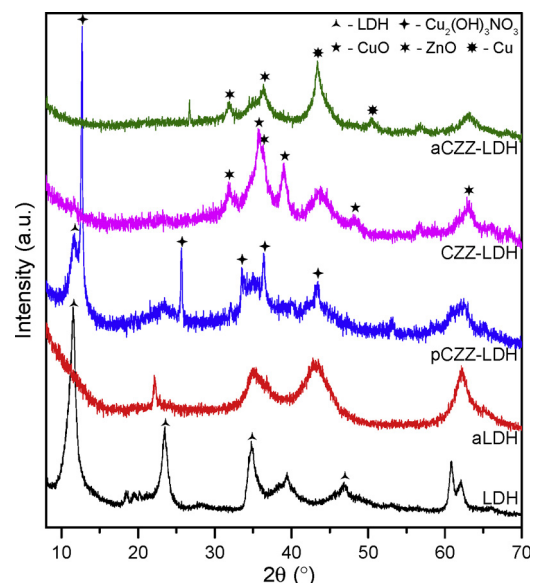


Fig. 2. XRD patterns of LDH, aLDH, pCZZ-LDH, CZZ-LDH and aCZZ-LDH.

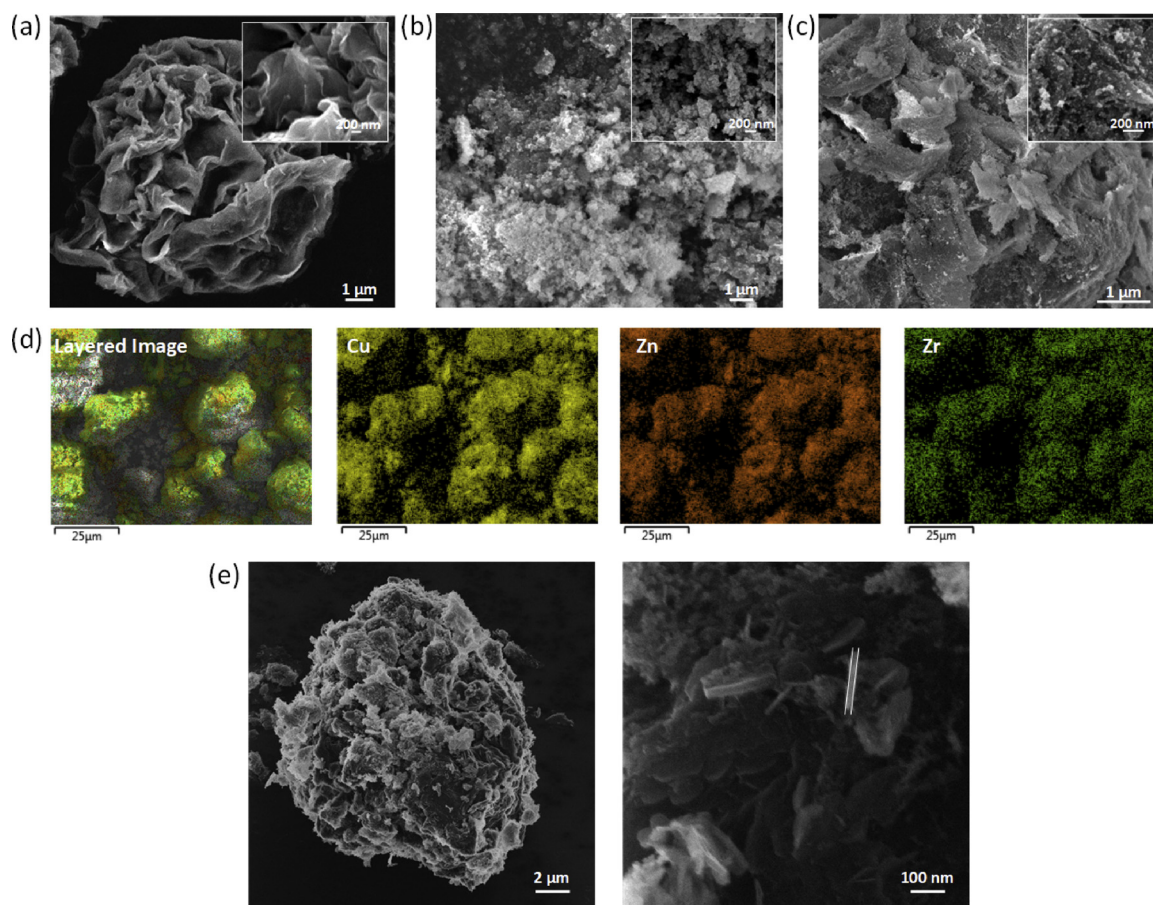


Fig. 3. SEM images of (a) aLDH, (b) CZZ and (c) CZZ-LDH. (d) EDX images of CZZ-LDH. (e) HIM images of aCZZ-LDH.

successfully obtained. However, peaks related to zirconium are not observed in these samples due to its low content.

SEM images of as-obtained aLDH, CZZ and CZZ-LDH are shown in Fig. 3. The aLDH (Fig. 3(a)) shows a hierarchical “desert rose” morphology on which inter-grown platelets are observed [36]. From the SEM image of CZZ (Fig. 3(b)), it appears as spheres on a nano scale. Due to the absence of support, these spheres do not agglomerate, but tend to accumulate, restricting copper dispersion. From the morphology of CZZ-LDH sample (Fig. 3(c)), however, the outlook image is similar to that of aLDH at low magnification, and nanoparticles of CuO-ZnO-ZrO₂ can be distinguished on the surface in the enlarged view. As expected, these nanoparticles neither agglomerate nor stack, demonstrating good copper dispersion. EDX mappings in Fig. 3(d) also supplement this conclusion from visual distributions of Cu, Zn and Zr.

Bulk and surface compositions obtained from ICP and EDX are compared in Table 1. The bulk aLDH sample contains magnesium and alumina with a Mg:Al mole ratio of 1.04:1 (ICP) and the surface of the sample with Mg:Al mole ratio of 1.3:1 (EDX), which is close to the report provided by the supplier that Mg:Al mole ratio is 0.9:1. The bulk

as-synthesized CZZ has a Cu:Zn:Zr mole ratio of approximately 7.3:3.9:1 (ICP) and the surface of the sample of 9.8:7.1:1 (EDX). In the heterogeneous catalyst of CZZ-LDH, the ratios of metal elements of Cu:Zn:Zr and Mg:Al should be similar to that of CZZ and LDH. The surface contents of Cu and Zn (EDX), however, are slightly higher than the bulk contents (ICP), verifying the presence of CZZ on the surface of the HT.

From HIM images in Fig. 3 (e), the high-resolution pictures of aCZZ-LDH shows that the aCZZ-LDH inherited LDH lamellas. From Fig. 3 (a), (c) and (e), aLDH, CZZ-LDH and aCZZ-LDH samples all exhibit porous structures and retain the LDH lamella morphology, which further confirms the existing LDH frameworks even in aCZZ-LDH.

3.1.2. Surface areas and copper dispersion

BET surface areas of the synthesized samples are summarized in Table 2. After activation at 673 K, aLDH shows a relatively high BET surface area of 195.2 m² g⁻¹, which may enhance the dispersion of catalyst particles. BET surface areas of CZZ and aCZZ are quite low (around 35 m² g⁻¹), which is consistent with those of reported metal nanoparticles [8,37]. When 40.7 wt.% nonporous particles of CuO-ZnO-ZrO₂ was loaded on the surface of aLDH, the surface area of CZZ-LDH was reduced to 152.3 m² g⁻¹, but it is still 4.3 times larger than that of CZZ (35.7 m² g⁻¹). After reduction, both surface areas of aCZZ-LDH and aCZZ slightly decrease from 152.3 to 136.0 and 35.7 to 33.8 m² g⁻¹, respectively, because the generated new phase of Cu⁰ sinters upon exposure to a reducing environment [38].

Pore size distributions also change before and after CZZ is loaded on LDH. The volume of large pores (50–100 nm) decreases significantly and those of medium (2–50 nm) and micro ones increase, which indicates that CZZ are mainly anchored on LDH external surface but does not enter its interlayers (< 0.8 nm from XRD). The same conclusion can

Table 1

Chemical compositions of samples (wt.%).

	Bulk composition ^a					Surface composition ^b				
	Mg	Al	Cu	Zn	Zr	Mg	Al	Cu	Zn	Zr
aLDH	26.9	29.2	n.a.	n.a.	n.a.	24.3	21.2	n.a.	n.a.	n.a.
CZZ	n.a.	n.a.	45.7	24.5	9.0	n.a.	n.a.	42.5	31.4	6.2
CZZ-LDH	15.0	21.2	12.2	9.6	5.7	16.0	15.9	18.5	11.0	3.3

^a Obtained from ICP.

^b Obtained from EDX mappings.

Table 2
Surface areas (SA) and copper dispersion of synthesized samples.

	BET surface area (m ² g ⁻¹) ^a	Pore volume (10 ⁻² cm ³ g ⁻¹) ^a				Cu surface area (m ² g ⁻¹) ^b	Cu surface area (m ² g _{Cu} ⁻¹) ^b	Cu dispersion (%) ^b
		< 2 nm	2–50 nm	50–100 nm	Total			
aLDH	195.2	1.8	10.4	3.6	15.8	n.a.	n.a.	n.a.
CZZ	35.7	0.7	17.4	10.8	28.9	n.a.	n.a.	n.a.
aCZZ	33.8	0.6	24.1	12.7	37.4	39.3	86.0	22.9
CZZ-LDH	152.3	2.6	21.5	3.2	27.3	n.a.	n.a.	n.a.
aCZZ-LDH	136.0	2.7	20.4	3.2	26.3	29.9	245.1	66.0

^a Calculated by N₂ isotherms at 77 K (Figure S2).

^b Calculated by N₂O titration.

also be drawn from SEM images (Fig. 3(c)). This formation results in diminished interlayer space of LDH, which has been confirmed by XRD, and developed mesopores and micropores of CZZ-LDH and aCZZ-LDH indirectly.

The superior active site presentation of aCZZ-LDH was demonstrated by surface copper and dispersion as summarized in Table 2. The copper surface area of aCZZ is 39.3 m² g⁻¹ (slightly exceeds the BET surface area due to different measuring method [39]), yet that of aCZZ-LDH which only contains ~38 wt.% active component attains 29.9 m² g⁻¹. It can be clearly seen that copper surface area per gram copper reaches 86.0 m² g_{Cu}⁻¹ in the aCZZ sample but only 245.1 m² g_{Cu}⁻¹ in aCZZ-LDH samples. The copper dispersion again confirms the efficient utilization of copper, giving 66.0% dispersion in aCZZ-LDH, whereas the value is only 22.9% in aCZZ.

3.1.3. Synthesis scheme

Based on the analysis above, the schematic of the CZZ-LDH evolution involves 4 steps as shown schematically in Fig. 4: (1) crystal LDH is activated by calcination at 673 K, where anions (viz., OH⁻ and CO₃²⁻) in the interlayers are partly decomposed [40] and an amorphous structure LDH is formed, the increase of interlayer spacing resulting in a large surface area; (2) co-precipitation on the surface of the activated LDH is completed at a pH of 9.5, and a “memory effect” of LDH helps the pCZZ-LDH to restructure; (3) in the following calcination at 673 K, the precipitates are thermally decomposed to form metal oxides and the LDH is reactivated (CZZ-LDH); (4) the final catalyst is obtained by reduction under hydrogen flow at 573 K, where the copper oxide in the sample is converted into Cu⁰ (aCZZ-LDH).

3.2. Catalytic reaction experiments

In order to investigate the impact of LDH on the catalytic reaction, the prepared catalyst (aCZZ-LDH) was tested for CO₂ hydrogenation and also compared with a reference catalyst within a wide temperature range from 423 to 623 K at 3.0 MPa. Two overall reactions (see reaction equation (6–7)) occur simultaneously during CO₂ hydrogenation. From the thermodynamic point of view, CO₂ hydrogenation to methanol is exothermic and converts from 4 to 2 mol but to CO is endothermic with no change in mole number, thus, methanol synthesis should be favoured at low temperature and high pressure.

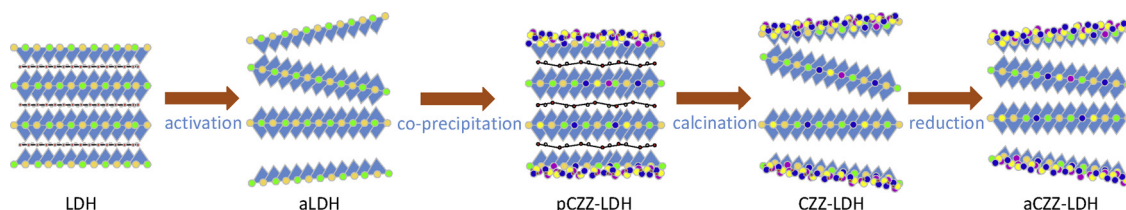
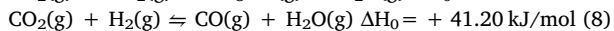
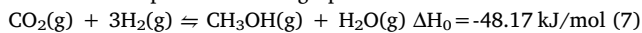


Fig. 4. Schematic of CZZ-LDH synthetic processes.

It was found that catalytic performances under GHSVs of ~2000 and ~3000 h⁻¹ were fairly close ($X_{\text{CO}_2} = 4.9$ and 4.8%, $S_{\text{MeOH}} = 78.3$ and 82.5%, respectively), thus, following reactions were all conducted under the GHSV of ~2000 h⁻¹.

3.2.1. LDH enhanced catalytic performances

CO₂ hydrogenation reaction was first conducted in the fixed bed reactor using the prepared catalysts, i.e., aCZZ-LDH, aCZZ and aLDH, and the experimental results are presented in Table 3. Because the substrates containing carbon element detected by GC in the system (both in gas and liquid phases) are only methanol, CO₂, CO and CH₄, the carbon balance can be calculated according to the amounts of these four chemicals (always within 100 ± 7% in repeating experiments). The aLDH sample is a control sample and confirmed that no catalytic function is provided by LDH alone. The reaction performance using aCZZ-LDH catalyst shows improved methanol selectivity with a lower CO₂ conversion than aCZZ due to lower metal content. At the given high temperature of 526 K and pressure of 3.0 MPa, the methanol selectivity with aCZZ-LDH is 78.3% but the selectivity with aCZZ is only 59.5%. Such high methanol selectivity is unreachable for conventional copper-based catalysts without further modification. As shown in Table 3, the methanol selectivity for other similar catalysts reported previously (Cu-ZnO/ZrO₂, C6Z3Z1-CB and C6Z3Z1-OX) is only around 50%. Theoretically, both methanol synthesis (MS, equation (7)) and reverse water gas shift (RWGS, equation (8)) reactions take place on the surface of active copper. As the reaction temperature increases, the CO by-product will inevitably be formed and easily accounted for over 50% of total products when the temperature exceeds 523 K at 3.0 MPa (shown in Fig. 6). In this work, however, the aCZZ-LDH successfully breaks that trend under a reasonable operating pressure, achieving remarkable methanol selectivity.

The CO₂ conversion and STY_{total} based on mass of total catalyst with aCZZ-LDH is lower than that with aCZZ. However, the STY based on the mass of active component and copper with aCZZ-LDH is much higher (96.0 mg_{MeOH} g_{active component}⁻¹ h⁻¹ and 299.0 mg_{MeOH} g_{Cu}⁻¹ h⁻¹), which indicates the benefits from the improved copper surface area and better copper dispersion. Accordingly, the efficiency of copper utilization is significantly enhanced with the LDH support, so that the novel catalyst exhibits a competitive edge even compared with some novel catalysts (shown in Table 3).

Moreover, the considerable lifetime is promising for the aCZZ-LDH catalyst to be commercialised (Table S2 in SI). The CO₂ conversion with

Table 3

Catalytic performances of as-obtained samples and some reported catalysts.

	Reaction Condition		X_{CO_2} (%)	S_{MeOH} (%)	$\text{STY}_{\text{total}}$ (mg _{MeOH} g _{cat.} ⁻¹ h ⁻¹)	$\text{STY}_{\text{active component}}$ (mg _{MeOH} g _{active component} ⁻¹ h ⁻¹)	STY_{Cu} (mg _{MeOH} g _{Cu} ⁻¹ h ⁻¹)
	Temperature (K)	Pressure (Mpa)					
aCZZ-LDH	523	3.0	4.9	78.3	36.5	96.0	299.0
aCZZ	523	3.0	7.5	59.5	48.3	48.3	105.7
aLDH	523	3.0	0	0	0	0	0
C-CZA	523	3.0	6.33	68.6	48.9	48.9	94.7
Cu-ZnO/ZrO ₂ [30]	513	3.0	17.0	41.5	~48.8	~48.8	~121.7
1.0CZZ-cb [41]	513	3.0	15.0	31.0	700	700	3370.7
1.0CZZ-cp [41]	513	3.0	14.0	13.0	320	320	1400.8
CuZnZr [42]	513	3.0	19.0	50.0	303	303	878.1
C6Z3Z1-CB [38]	513	3.0	16.0	48.7	n.a.	n.a.	n.a.
C6Z3Z1-OX [38]	513	3.0	18.0	51.2	305-1200	305-1200	672~2645
Cu@ZnO [43]	523	3.0	2.3	100	147.2	147.2	241.5
Pd/CeO ₂ [44]	533	3.0	5.2	84.7	n.a.	n.a.	n.a.
La-Cu/ZrO ₂ [45]	493	3.0	6.2	66	n.a.	n.a.	n.a.
Ga ₂ O ₃ -Pd/SiO ₂ [46]	523	3.0	1.34	58.9	283.4	283.4	n.a.
YBa ₂ Cu ₃ O ₇ [47]	513	3.0	3.4	50.7	n.a.	n.a.	n.a.

the aCZZ-LDH slightly decreases by 4.1% but the methanol selectivity remains unchanged after 96 h of continuous reactions at 523 K. The catalytic characteristics can be well recovered by H₂ reduction, showing 6.1% higher CO₂ conversion than the used samples above.

3.2.2. Comparison of catalytic properties between aCZZ-LDH and a commercial catalyst

The reaction performance of the synthesized catalyst was investigated at various reaction temperatures (reaction pressure was maintained at 3 MPa and GHSV around 2000 h⁻¹ for all experiments) and then compared with those of a commercial copper-based catalyst (labelled as C-CZA, Afar Aesar) which is commonly used in methanol synthesis from syngas hydrogenation. The compositions and properties of the commercial catalyst as shown in Table S1 shows higher copper surface area but three times lower copper dispersion, compared with the synthesized aCZZ-LDH (in Table 2).

Fig. 5 shows that both catalysts present a similar trend as temperature increases, however, C-CZA contains more than double the amount of active copper than aCZZ-LDH per unit weight so that the C-CZA always shows higher CO₂ conversion over the whole measured temperature ranges. Herein, both reaction rates (equations 7 and 8) are accelerated at a high temperature but RWGS is significantly promoted due to its endothermic nature, mainly contributing to enhanced CO₂ conversion.

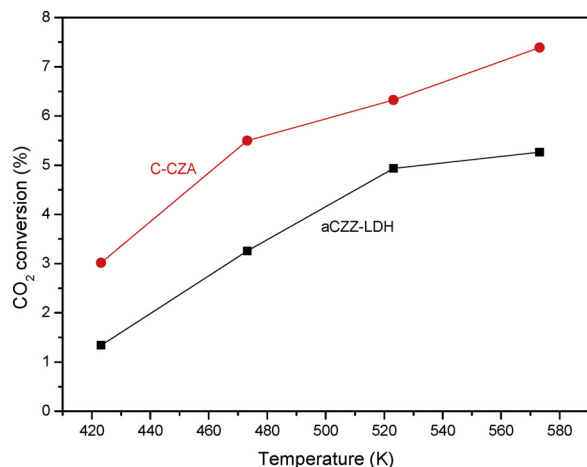


Fig. 5. CO₂ conversion using the prepared and commercial catalysts at various reaction temperatures (Pressure: 3.0 MPa, GHSV: ~2000 h⁻¹).

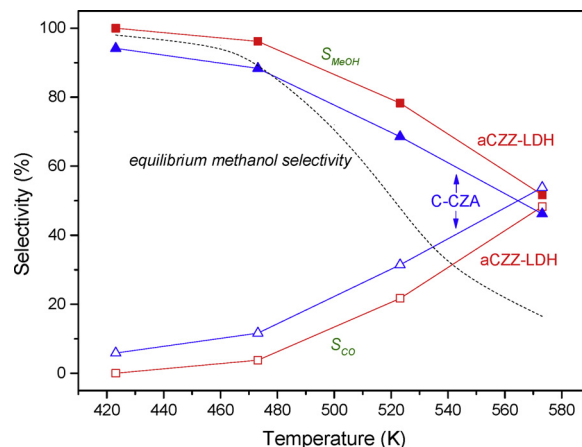


Fig. 6. Selectivity of methanol and carbon monoxide at various reaction temperatures (Pressure: 3.0 MPa, GHSV: ~2000 h⁻¹).

Because methanol synthesis is exothermic, the trend of methanol selectivity decreases and is just opposite with that of CO for both aCZZ-LDH and C-CZA as the reaction temperature increases (Fig. 6). When the temperature is lower than 473 K, the methanol selectivity for both aCZZ-LDH and C-CZA are ~90% or higher as the RWGS reaction is much reduced with these catalysts. When the reaction temperature increases to 523 K, however, this reaction is promoted to generate CO so that methanol selectivity decreases simultaneously. Compared with C-CZA, the prepared aCZZ-LDH presents higher methanol selectivity in the measured temperature range, which we attribute at least partly to the enhanced CO₂ adsorption characteristics of LDH. Because of CO₂ adsorption on LDH-derived amorphous oxide in the system, CO₂ concentration around active sites of CZZ is raised to form a “CO₂ rich region”, which is equivalent to increase the CO₂ partial pressure in the reaction system. Therefore, the real partial pressure of CO₂ near active copper sites is higher than the CO₂ pressure in the system, contributing to elevated methanol selectivity above the calculated equilibrium level. Although the commercial C-CZA shows considerable methanol selectivity due to the fine-tuned metal interfaces, the aCZZ-LDH exhibits further developed methanol selectivity profited from its improvements in CO₂ adsorption which enforces the methanol synthesis.

The space time yields ($\text{STY}_{\text{total}}$, $\text{STY}_{\text{active component}}$ and STY_{Cu}) were further investigated and are summarized in Fig. 7. The methanol $\text{STY}_{\text{total}}$ of aCZZ-LDH and C-CZA exhibits a bell shape as the reaction temperature increases and achieves a peak at 523 K, which results from

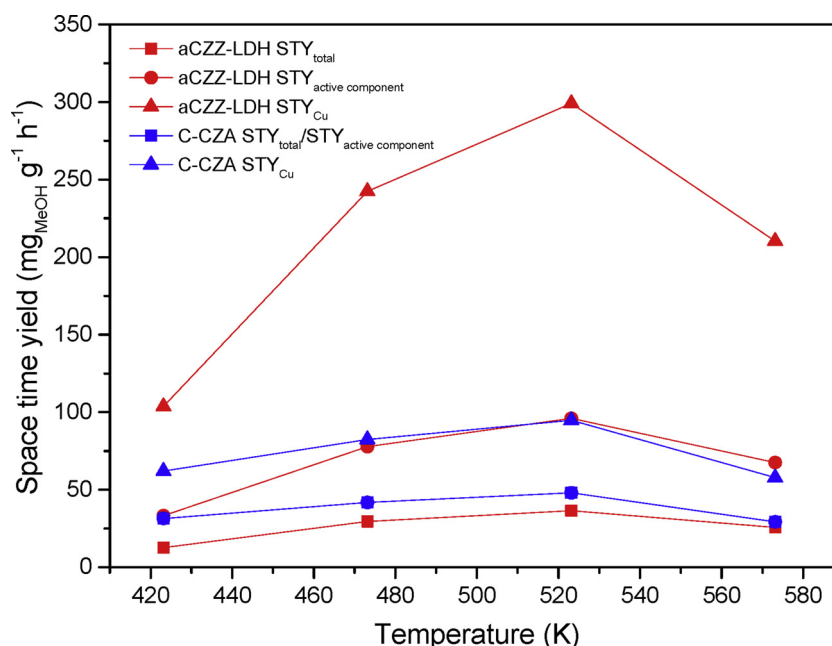


Fig. 7. Comparisons of STY_{total}, STY_{active component} and STY_{Cu} of CZZ-LDH and C-CZA (Pressure: 3.0 MPa, GHSV: ~2000 h⁻¹).

competitions of MS and RWGS reactions. Thus, 523 K is usually chosen as the operating temperature for CO₂ or syngas hydrogenation [48]. Meanwhile, the STY_{total} for C-CZA is higher than that of aCZZ-LDH because there is more active metal per unit catalyst in C-CZA. However, on the basis of active metal and copper content, STY_{active component} and STY_{Cu} of aCZZ-LDH exceed those of C-CZA. At the experimental condition of 523 K and 3.0 MPa, for instance, aCZZ-LDH shows twice the STY_{active component} and 3.16 times the STY_{Cu} than C-CZA, which further confirms the advantages of LDH in the prepared catalyst.

4. Conclusions

A heterogeneous catalyst based on Cu-ZnO-ZrO₂ supported on LDH (denoted CZZ-LDH) was synthesized via an improved co-precipitation method in this investigation. The synthesized catalyst CZZ-LDH presented higher BET surface area of 152.3 m² g⁻¹ and a greater copper dispersion of 66.0%, compared to the one without LDH. From CO₂ hydrogenation reactions in a fixed-bed reactor, the catalyst showed higher methanol selectivity than conventional copper-based catalysts. The CZZ-LDH, moreover, exhibited doubled the space time yield based on active metal (STY_{active}) than a commercial methanol synthesis catalyst at mild conditions of 523 K and 3.0 MPa with a remarkable methanol selectivity of 78.3%. Therefore, the utilization efficiency of copper was visibly elevated in CZZ-LDH. We hypothesize that the improved performance may be because of a high adsorption capacity of CO₂ on LDH-derived amorphous oxide, increasing the adsorbed concentration of CO₂ near the active metal sites.

Acknowledgements

The authors gratefully acknowledge the financial support of the China Scholarship Council (No. 201606080053), the University of Melbourne and the Natural Science Foundation of China (Grants 51474067). We acknowledge the Monash X-ray platform (MXP) for XRD measurements and Monash centre of electron microscopy (MCEM) for SEM-EDX analyses. We also acknowledge the Materials Characterisation and Fabrication Platform (MCFP) in the University of Melbourne and the Victorian Node of the Australian National Fabrication Facility (ANFF) for the HIM characterizations.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jcou.2018.11.006>.

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